

NP-completeness of aggregate root-dependent spanning-tree cost

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Abstract

An undirected spanning tree induces an arborescence at every vertex root. Given an independent arc-cost vector for each root, we prove that deciding whether the sum of the costs of these entire rooted orientations is at most a given threshold is NP-complete, even when every cost belongs to $\{0, 1, 2\}$. A reduction from satisfiability uses two hubs, variable vertices, and clause vertices. A symmetric cost vector at one hub forces a tree at threshold to encode an assignment with one selected literal per clause; the corresponding clause-root costs test those literals. Uniformly shifting the costs yields the same conclusion for costs in $\{1, 2, 3\}$. This resolves Problem 1 in Volker Kaibel’s November 2018 Oberwolfach workshop contribution, whose report was published in December 2019. Exact finite checks accompany the all-size proof.

1 Problem and result

Let $G = (V, E)$ be a finite connected simple undirected graph with $N = |V| \geq 1$, and let $A = \{(u, v), (v, u) : \{u, v\} \in E\}$. For every $r \in V$, the input supplies an independent cost vector $c_r : A \rightarrow \mathbb{Z}_{\geq 0}$. All $2N|E|$ entries are explicitly encoded in binary. For a spanning tree T , let T^r denote its entire edge set oriented away from r , and define

$$F(T) = \sum_{r \in V} \sum_{(u,v) \in T^r} c_r(u, v). \quad (1)$$

The decision question is whether $F(T) \leq K$ for some spanning tree, where $K \in \mathbb{Z}$ is part of the input. The same common undirected tree is used at every root; the costs charge all its oriented edges. Reversing every input cost, $c'_r(u, v) = c_r(v, u)$, gives the equivalent inward-orientation convention.

Kaibel’s Problem 1 asks for the hardness of minimizing precisely this all-root objective with real arc-cost vectors [1]. The nonnegative integral subclass below therefore answers that optimization question under ordinary finite input encoding. Kaibel motivates the question by integer optimization over Martin’s extension of the spanning-tree polytope. That formulation is classical, as described in the inspected expositions of Conforti, Cornuéjols and Zambelli [3, Section 6.1] and Friesen [4, Proposition 2]: common edge variables satisfy $z_r(u, v) + z_r(v, u) = x_{\{u,v\}}$, with $z \geq 0$, root indegree zero and all other indegrees one. For integral common edge variables these constraints select one tree, and its root orientations are determined uniquely. Integrality of the projection onto the edge variables does not establish integrality of the full joint lift under arbitrary objectives on all z variables. The formulation and this interpretation are not contributions of this note.

Theorem 1. *The decision problem for (1) is NP-complete even when every cost belongs to $\{0, 1, 2\}$. Consequently it is strongly NP-complete, and finding a minimum-cost tree is NP-hard.*

Membership in NP follows by checking that the proposed $N - 1$ edges form a spanning tree, traversing it from each root, and summing the supplied binary costs. The number of operations and the lengths of intermediate integers are polynomial in the explicit input size.

2 Reduction

We reduce from satisfiability of a conjunctive normal form whose clauses have at most three literals, using the standard NP-completeness result for this restriction [2]. Remove repeated literals within each clause and discard tautological clauses. An empty clause gives a fixed no-instance: two vertices joined by one edge, all four root-arc costs equal to 1, and threshold 1. Its unique tree has aggregate cost 2. If no clauses remain, use the same graph with all costs zero and threshold zero, a fixed yes-instance. Delete unused variable declarations and relabel the variables that actually occur densely. Thus n is the number of occurring variables, regardless of their original labels. It remains to consider $n \geq 1$ variables $x_1, \dots, x_n$ and $m \geq 1$ nonempty clauses $C_1, \dots, C_m$, each with at most three distinct variables and without both signs of a variable in one clause.

The graph has two hubs t, f , a vertex v_i for every variable, and a vertex q_j for every clause. Its edges are

$$\{t, f\}, \quad \{t, v_i\}, \{f, v_i\} \ (1 \leq i \leq n), \quad \{q_j, v_i\} \text{ when a literal on } x_i \text{ occurs in } C_j.$$

It is connected and simple, with $N = n + m + 2$ vertices and at most $1 + 2n + 3m$ edges. Set

$$K = 2m + n. \tag{2}$$

At root t , assign the same cost to both orientations of each edge:

$$\begin{aligned} c_t(t, f) &= c_t(f, t) = 0, \\ c_t(t, v_i) &= c_t(v_i, t) = c_t(f, v_i) = c_t(v_i, f) = 1, \\ c_t(q_j, v_i) &= c_t(v_i, q_j) = 2. \end{aligned} \tag{3}$$

At root q_j , the only potentially nonzero costs are variable-to-hub arcs for variables occurring in C_j :

literal in C_j	$c_{q_j}(v_i, t)$	$c_{q_j}(v_i, f)$
x_i	0	1
$\neg x_i$	1	0

(4)

All other costs at q_j are zero, including the reverse hub-to-variable arcs. The vectors at f and all variable roots are identically zero. These prescriptions specify every entry of the input table, without using any tree-dependent data.

Lemma 1 (Structural threshold). *If a spanning tree has $F(T) \leq K$, it contains the hub edge, every clause vertex is a leaf, and every variable has exactly one hub attachment.*

Proof. Let q be the number of clause-variable edges in T , p the number of hub-variable edges, and $h \in \{0, 1\}$ the indicator of the hub edge. Each clause vertex has positive degree and is incident only to clause-variable edges, each of which meets exactly one clause vertex. Hence $q \geq m$. Counting all edges in the tree gives $p + q + h = n + m + 1$. By the symmetric costs at t ,

$$F_t(T) := \sum_{a \in T^t} c_t(a) = p + 2q = K + (1 - h) + (q - m). \tag{5}$$

All other root costs are nonnegative. The two terms after K are nonnegative integers, so $F(T) \leq K$ forces $h = 1$ and $q = m$. Every clause vertex is therefore a leaf. Deleting these leaves leaves a tree on the hubs and variables containing the hub edge. Each variable must meet at least one hub, and meeting both would create a triangle with the hub edge. Each meets exactly one. $\square$

Call a tree of this form structured. Encode an assignment by making x_i true exactly when $\{t, v_i\}$ is present. Each clause leaf selects its unique neighboring variable and hence its unique literal in that clause. Conversely, any assignment together with one selected literal in each clause gives a structured tree, with $F_t(T) = K$.

Lemma 2 (Literal costs). *For every structured tree,*

$$F(T) = K + \#\{j : \text{the literal selected by } q_j \text{ is false}\}. \quad (6)$$

Proof. Root the structured tree at q_j , and let v_i be its unique neighbor. The path into the hub core begins $q_j \rightarrow v_i \rightarrow h_i$, where h_i is the selected hub of v_i . Thus this variable's hub edge is directed from the variable to its hub. Every other variable's hub edge is directed from its hub toward that variable: the core consists of two stars joined by the hub edge, and the route from q_j enters every other variable from the hub side. Accordingly, the only arc that can have nonzero cost at root q_j is $v_i \rightarrow h_i$. Table (4) assigns this arc cost zero precisely when the selected literal is true, and one otherwise. All clause edges, the hub edge, and the remaining root vectors contribute zero beyond the base cost. Summing over the clause roots proves the identity. $\square$

If the formula is satisfiable, attach every variable to its truth-value hub and attach every clause leaf to a true literal. The resulting tree has cost K . Conversely, a tree of cost at most K is structured by Lemma 1; Lemma 2 makes each selected literal true, so every clause is satisfied. This proves the equivalence of the two decision instances.

The graph, all N cost vectors, and the threshold are computable in polynomial time. There are $2N|E|$ explicitly supplied entries, each in $\{0, 1, 2\}$, and $K = 2m + n \leq 2N$. This completes the proof of Theorem 1, including hardness under unary numerical encoding.

Corollary 1. *The same decision problem is strongly NP-complete when every cost belongs to $\{1, 2, 3\}$.*

Proof. Add one to every arc cost at every root and replace K by $K + N(N - 1)$. Every spanning tree has $N - 1$ arcs at each of the N roots, so every objective changes by exactly this same constant. $\square$

The original audited choice placed $B = n + 1$ on clause-variable edges. For that same construction at any integer $B \geq 2$, with $K_B = Bm + n$, the unrestricted base identity is

$$F_t(T) = p + Bq = K_B + (1 - h) + (B - 1)(q - m).$$

The choice $B = 2$ above is a consequence of that identity and the same literal test. The transpose and positive shift are likewise consequences of the same result. Equation (6) is asserted only for structured trees; it does not describe the unrestricted global optimum by the minimum number of unsatisfied clauses.

3 Verification, attribution and status

The accompanying standard-library Python package constructs full explicit cost tables and enumerates all spanning trees for small test instances. It sums costs by independently orienting each whole tree from every root and also by partitioning roots across each edge cut. It checks the unrestricted base identity, individual clause-root costs, both reduction directions, inward transposition, uniform positive shifts, normalization with sparse variable labels, and trivial cases. Deliberately false guards, corrupted costs and an incorrect proof binding are rejected under normal Python and Python -0. Finite computations supplement the written all-size proof; they do not establish NP-hardness or historical priority.

The question is attributed to Kaibel’s workshop contribution of 4–10 November 2018, printed on pages 3014–3015 of the report published 17 December 2019 [1]. No earlier proof of the exact objective was located in the source and model audit completed on 6 October 2026. That dated audit has incomplete historical coverage and does not certify absolute first discovery or continued open status in 2026. Its source scopes and access limitations are recorded in the supporting provenance. The result concerns this exact whole-orientation objective; no claim is made here about Kaibel’s neighboring Problem 2.

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